DEPI PROJECT DOCUMENTATION

Requirements Gathering

Stakeholders • User Stories • Functional and Non-Functional Requirements • Testing

1. Project Overview

We are developing Churn Radar as a customer-retention analytics project. We use Python for data preparation, exploratory analysis and modelling; SQL for KPI calculations; Power BI for interactive dashboards; and Streamlit to demonstrate customer risk scores.

2. User Stories

ID	User Story	Priority	Acceptance Criteria
US-01	View overall churn rate	Must	The dashboard shows customer count, churn count and churn rate for all rows with a valid Churn label.
US-02	Compare customer segments	Must	The dashboard shows counts and churn rates for at least three customer segments.
US-03	Check data quality	Must	Our analysis reports data types, missing values, duplicate rows and the Churn distribution.
US-04	Verify KPIs with SQL	Must	SQL and Python counts match exactly, and rates match to one decimal place.
US-05	Compare churn models	Must	We evaluate Logistic Regression and Random Forest using the same test set.
US-06	Understand model errors	Should	We report false positives, false negatives and the selected classification threshold.
US-07	Filter dashboard results	Should	At least two dashboard filters update the relevant visuals.
US-08	Rerun the project	Must	Our README documents the dataset source, environment, dependencies and run steps.

3. Use Cases

Use Case ID	Actor	Main Action	Expected Result
UC-01	Business user	Reviews headline dashboard metrics.	Understands overall churn.
UC-02	Business user	Selects a segment and compares its churn rate.	Identifies groups for further review.
UC-03	Data analyst	Loads, inspects and cleans the CSV, then runs SQL checks.	Produces consistent data and KPI results.
UC-04	Model reviewer	Compares model metrics and prediction errors.	Understands model performance.
UC-05	Dashboard user	Applies filters and reviews charts.	Explores results interactively.
UC-06	DEPI evaluator	Reviews project files, results and test evidence.	Checks project completeness.

4. Functional Requirements

ID	Requirement	Priority	Acceptance Check
FR-01	Load the selected IBM Telco Customer Churn CSV (21-column version).	Must	The loading check confirms 7,043 rows and 21 columns; a schema mismatch produces a clear error.
FR-02	Profile data quality.	Must	Our data-quality report includes data types, missing values, duplicates and Churn distribution.
FR-03	Clean TotalCharges and prepare predictors.	Must	We convert TotalCharges to numeric, replace blank values with 0 when tenure is 0, retain those rows and record replacements in the cleaning log.
FR-04	Calculate KPIs in SQLite.	Must	Our SQL queries calculate total customers, churn count, churn rate and at least three segment rates.
FR-05	Perform exploratory and statistical analysis.	Must	Our analysis answers at least two project questions and explains the results.
FR-06	Train and compare models.	Must	We compare Logistic Regression and Random Forest using the same data split.
FR-07	Evaluate prediction quality.	Must	We report the confusion matrix, precision, recall, F1 and ROC-AUC. Our targets are ROC-AUC ≥ 0.75 and recall ≥ 0.70; we select the threshold using validation data and report the trade-offs.
FR-08	Check class imbalance.	Must	We report class proportions and compare class weighting with the baseline. We use SMOTE only if needed and only on training data.
FR-09	Build Power BI dashboard.	Must	Our dashboard includes an overview and three segment views. Counts match SQL exactly and rates match to one decimal place.
FR-10	Explain errors at the selected threshold.	Should	We report false positives and false negatives and explain how we selected the threshold.
FR-11	Explain important features.	Could	We present feature importance or SHAP results if this optional analysis is included.

5. Non-Functional Requirements

ID	Requirement	Priority	Test ID
NFR-01	The project can be rerun using the README.	Must	T-07
NFR-02	Missing required columns produce a clear error.	Must	T-01
NFR-03	The raw CSV remains unchanged and cleaning is logged.	Must	T-02
NFR-04	The repository contains no credentials or private customer-level data.	Must	T-09
NFR-05	Dashboard labels and KPI definitions are clear.	Should	T-06
NFR-06	The core notebook runs in under 5 minutes on the agreed machine.	Must	T-07
NFR-07	Code is organized into clear analysis stages.	Should	T-10
NFR-08	Every Must requirement has a linked test and evidence.	Must	T-08
NFR-09	The dashboard's initial page loads in under 5 seconds.	Must	T-06
NFR-10	Across SQL, Python and Power BI, counts match exactly and rates match to one decimal place.	Must	T-03, T-06

6. Test Plan & Traceability

Test ID	Test / Evidence	Owner
T-01	We check the schema, row/column count and missing-column error.	Nada
T-02	We check TotalCharges conversion, row counts and the cleaning log.	Nada + Mohamed
T-03	We compare SQL and Python counts exactly and rates to one decimal place.	Mohamed
T-04	We review findings from at least two statistical questions.	Mohamed + Nada
T-05	We review model metrics, threshold, class balance and held-out test set.	Abdelrahman
T-06	We check dashboard filters, labels, load time, counts and rates.	Catherine + Mohamed
T-07	We rerun the project using the README and record notebook runtime.	Nada
T-08	We audit whether every Must requirement has a test and evidence.	Judy
T-09	We scan the repository for credentials and private data.	Judy
T-10	We review code stages: load, clean, explore, model, evaluate and export.	Abdelrahman

7. Dataset & Submission

We use the 21-column Telco Customer Churn CSV and keep the raw file unchanged. Our KPI denominator is every row with a valid Churn label. We convert blank TotalCharges values to 0 when tenure is 0 and retain those rows; the cleaning log records the replacements found in our file.

Submission: We upload all project documents, code, SQL queries, dashboard files and test evidence to the project GitHub repository.